

compare the coverage area of sub-6 ghz and mm wave frequencies using python.

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Sub:-5G fundamentals and architectures

Assignment – 03

1. Objective :-

The objective of this simulation is to compare the coverage area of Sub-6 GHz and mmWave frequencies using Python. The simulation analyzes the received signal power at different communication distances for a 3.5 GHz Sub-6 GHz signal and a 28 GHz mmWave signal.

The Free-Space Path Loss (FSPL) model is used to calculate the propagation loss at different distances. The maximum coverage distance is determined based on a minimum acceptable received power level. The coverage area is then calculated by assuming circular coverage.

2. Software and Tools Used :-

Software / Tool	Purpose
Python	Main programming language
NumPy	Numerical calculations
Matplotlib	Graph generation and visualization
Google Colab	Executing and testing Python code
Free-Space Path Loss Model	Estimating propagation loss

3. Simulation Methodology :-

The simulation follows these steps:

Step 1: Define Sub-6 GHz and mmWave Frequencies

Two different frequencies are selected for the comparison.

Sub-6 GHz frequency:

$$f = 3.5 \text{ GHz}$$

mmWave frequency:

$$f = 28 \text{ GHz}$$

The same transmitter power and antenna gains are used for both frequencies so that the effect of frequency on coverage can be compared.

Step 2: Define System Parameters

The following system parameters are considered for the simulation:

Transmit Power:

$$P_T = 30 \text{ dBm}$$

Transmitter Antenna Gain:

$$G_T = 10 \text{ dBi}$$

Receiver Antenna Gain:

$$G_R = 10 \text{ dBi}$$

Minimum Acceptable Received Power:

$$P_R(\text{min}) = -90 \text{ dBm}$$

The communication distance is varied from 10 m to 50 km to determine the approximate coverage range of both frequencies.

Step 3: Calculate Wavelength

The wavelength of an electromagnetic signal depends on its operating frequency.

The wavelength is calculated using:

$$\lambda = c / f$$

Where:

- λ = Wavelength in metres (m)
- c = Speed of light = 3×10^8 m/s
- f = Operating frequency in Hertz (Hz)

For Sub-6 GHz frequency:

$$f = 3.5 \text{ GHz}$$

$$f = 3.5 \times 10^9 \text{ Hz}$$

Therefore,

$$\lambda = (3 \times 10^8) / (3.5 \times 10^9)$$

$$\lambda \approx \mathbf{0.0857 \text{ m}}$$

Therefore, the wavelength at 3.5 GHz is approximately:

$$\lambda \approx \mathbf{8.57 \text{ cm}}$$

For mmWave frequency:

$$f = 28 \text{ GHz}$$

$$f = 28 \times 10^9 \text{ Hz}$$

Therefore,

$$\lambda = (3 \times 10^8) / (28 \times 10^9)$$

$$\lambda \approx \mathbf{0.0107 \text{ m}}$$

Therefore, the wavelength at 28 GHz is approximately:

$$\lambda \approx 1.07 \text{ cm}$$

Comparison:

The 28 GHz mmWave signal has a much shorter wavelength than the 3.5 GHz Sub-6 GHz signal. The wavelength is used in calculating the free-space path loss of the communication link.

Step 4: Calculate Free-Space Path Loss

The Free-Space Path Loss is calculated using:

$$\text{FSPL} = 20 \log_{10} (4\pi d / \lambda)$$

Where:

- FSPL = Free-Space Path Loss in dB
- d = Distance between transmitter and receiver in metres
- λ = Wavelength in metres
- π = Mathematical constant

The path loss increases as the communication distance increases.

For the same distance, the higher-frequency mmWave signal experiences greater free-space path loss than the Sub-6 GHz signal.

Step 5: Calculate Received Power

The received power is calculated using:

$$\text{PR} = \text{PT} + \text{GT} + \text{GR} - \text{FSPL}$$

Where:

- PR = Received power in dBm
- PT = Transmit power in dBm

- GT = Transmitter antenna gain in dBi
- GR = Receiver antenna gain in dBi
- FSPL = Free-space path loss in dB

As the distance increases, the free-space path loss increases and the received power decreases.

Step 6: Determine Coverage Distance

The minimum acceptable received power is selected as:

PR(min) = -90 dBm

The maximum distance at which the received power is greater than or equal to -90 dBm is considered the approximate coverage distance.

The Python program calculates this distance separately for:

- 3.5 GHz Sub-6 GHz
- 28 GHz mmWave

Step 7: Calculate Coverage Area

The coverage area is assumed to be circular.

The area of a circle is calculated using:

$$A = \pi r^2$$

Where:

- A = Coverage area in km²
- r = Coverage radius in km

The calculated coverage distances are converted from metres to kilometres before calculating the coverage area.

4. Python Implementation :-

```
import numpy as np
import matplotlib.pyplot as plt

c = 3e8
f_sub6 = 3.5e9
f_mmwave = 28e9
Pt = 30
Gt = 10
Gr = 10
Pr_min = -90
distance_m = np.linspace(10, 50000, 3000)
def calculate_fspl(frequency, distance):
    wavelength = c / frequency
    fspl = 20 * np.log10((4 * np.pi * distance) / wavelength)
    return fspl
fspl_sub6 = calculate_fspl(f_sub6, distance_m)
fspl_mmwave = calculate_fspl(f_mmwave, distance_m)
Pr_sub6 = Pt + Gt + Gr - fspl_sub6
Pr_mmwave = Pt + Gt + Gr - fspl_mmwave
valid_sub6 = distance_m[Pr_sub6 >= Pr_min]
valid_mmwave = distance_m[Pr_mmwave >= Pr_min]
coverage_sub6 = valid_sub6[-1]
```

```

coverage_mmwave = valid_mmwave[-1]
area_sub6 = np.pi * (coverage_sub6 / 1000) ** 2
area_mmwave = np.pi * (coverage_mmwave / 1000) ** 2
print("Sub-6 GHz Frequency:", f_sub6 / 1e9, "GHz")
print("mmWave Frequency:", f_mmwave / 1e9, "GHz")
print("Minimum Received Power:", Pr_min, "dBm")
print("Sub-6 GHz Coverage Distance:", round(coverage_sub6 / 1000,
2), "km")
print("mmWave Coverage Distance:", round(coverage_mmwave /
1000, 2), "km")
print("Sub-6 GHz Coverage Area:", round(area_sub6, 2), "km2")
print("mmWave Coverage Area:", round(area_mmwave, 2), "km2")
print("Coverage Area Ratio:", round(area_sub6 / area_mmwave, 2))
plt.figure(figsize=(9, 6))
plt.plot(
    distance_m / 1000,
    Pr_sub6,
    linewidth=2,
    label="Sub-6 GHz (3.5 GHz)")
plt.plot(
    distance_m / 1000,
    Pr_mmwave,
    linewidth=2,
    label="mmWave (28 GHz)")

```

```
plt.axhline(
    Pr_min,
    linestyle="--",
    label="Minimum Required Power")
plt.xlabel("Distance (km)")
plt.ylabel("Received Power (dBm)")
plt.title("Received Power vs Distance")
plt.grid(True)
plt.legend()
plt.show()
plt.figure(figsize=(9, 6))
plt.plot(
    distance_m / 1000,
    fspl_sub6,
    linewidth=2,
    label="Sub-6 GHz (3.5 GHz)")
plt.plot(
    distance_m / 1000,
    fspl_mmwave,
    linewidth=2,
    label="mmWave (28 GHz)")
plt.xlabel("Distance (km)")
plt.ylabel("Free-Space Path Loss (dB)")
```

```
plt.title("Free-Space Path Loss vs Distance")
plt.grid(True)
plt.legend()
plt.show()

frequencies = [
    "Sub-6 GHz\n3.5 GHz",
    "mmWave\n28 GHz"]
areas = [
    area_sub6,
    area_mmwave]
plt.figure(figsize=(8, 6))
plt.bar(
    frequencies,
    areas)
plt.ylabel("Coverage Area (km2)")
plt.title("Coverage Area Comparison")
plt.grid(axis="y")
plt.show()
```

5. Output :-

Sub-6 GHz Frequency: 3.5 GHz

mmWave Frequency: 28.0 GHz

Minimum Received Power: -90 dBm

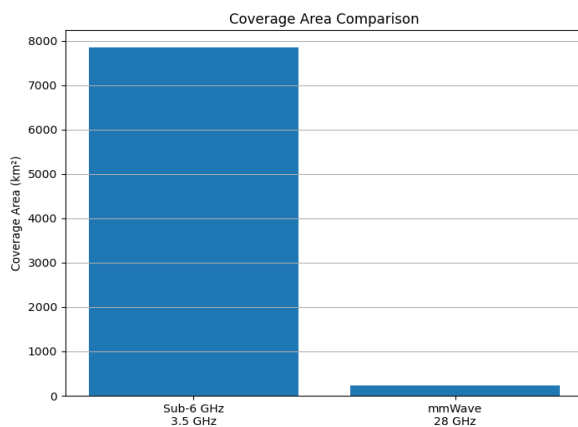
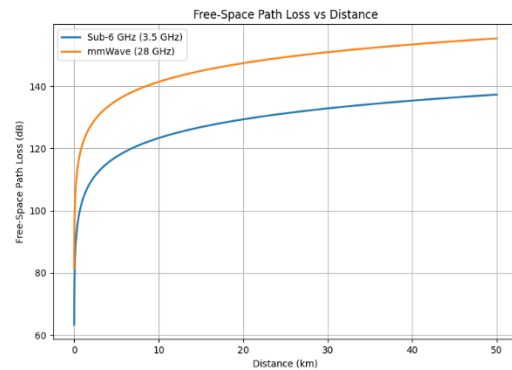
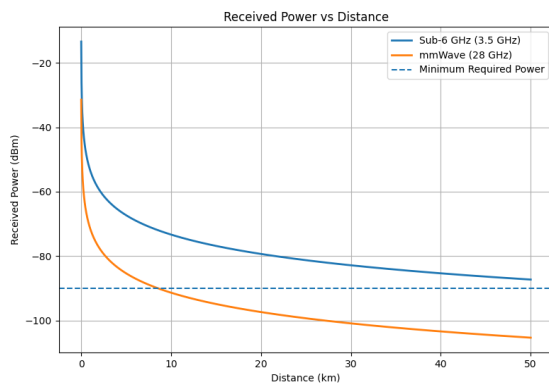
Sub-6 GHz Coverage Distance: 50.0 km

mmWave Coverage Distance: 8.51 km

Sub-6 GHz Coverage Area: 7853.98 km²

mmWave Coverage Area: 227.58 km²

Coverage Area Ratio: 34.51



6. Results :-

The Python simulation generates three graphs:

6.1 Received Power vs Distance

The graph shows the received power of Sub-6 GHz and mmWave signals as the communication distance increases.

The received power decreases with increasing distance because the free-space path loss increases.

The Sub-6 GHz signal maintains a higher received power over a longer distance compared with the 28 GHz mmWave signal.

Result:

Sub-6 GHz provides a larger coverage range than mmWave under the same system parameters.

6.2 Free-Space Path Loss vs Distance

The graph shows the variation of free-space path loss with communication distance.

Both frequencies experience increasing path loss as distance increases.

However, the 28 GHz mmWave frequency has greater path loss than the 3.5 GHz Sub-6 GHz frequency at the same distance.

Result:

The higher frequency experiences greater propagation loss, resulting in reduced coverage.

6.3 Coverage Area Comparison

The coverage area is calculated using:

$$A = \pi r^2$$

The simulation demonstrates that the coverage area of Sub-6 GHz is significantly larger than that of mmWave.

7. Result Table :-

Parameter	Sub-6 GHz	mmWave
Frequency	3.5 GHz	28 GHz
Approx. wavelength	0.0857 m	0.0107 m
Coverage distance	Approximately 50 km*	Approximately 8.5 km
Coverage area	Approximately 7854 km ²	Approximately 228 km ²
Propagation loss	Lower	Higher
Coverage	Larger	Smaller

Note: The Sub-6 GHz value reaches the upper simulation limit of 50 km. Therefore, 50 km should be interpreted as the maximum distance tested in this simulation rather than a universal real-world coverage limit.

8. Comparison of Sub-6 GHz and mmWave :-

Parameter	Sub-6 GHz	mmWave
Operating Frequency	3.5 GHz	28 GHz
Wavelength	Longer	Shorter
Free-Space Path Loss	Lower	Higher
Coverage Range	Larger	Smaller
Coverage Area	Larger	Smaller
Data Rate	High	Very High
Bandwidth	Moderate to High	Very High
Deployment	Wide-area coverage	Dense/localized coverage
Suitable Application	General 5G coverage	High-capacity hotspots

9. Result Analysis :-

The simulation demonstrates that frequency has a significant effect on wireless coverage.

The 3.5 GHz Sub-6 GHz frequency provides a larger coverage range compared with the 28 GHz mmWave frequency under the same transmit power and antenna gain conditions.

At 28 GHz, the wavelength is much shorter than at 3.5 GHz. As frequency increases, the free-space path loss also increases for a

given distance. This causes the received signal power to decrease more rapidly with distance.

Therefore, mmWave communication generally requires shorter distances between the transmitter and receiver and may require more base stations or access points for continuous coverage.

Sub-6 GHz frequencies are more suitable for broad-area coverage because they provide better propagation characteristics and larger coverage areas.

On the other hand, mmWave frequencies can provide very high bandwidth and data rates and are suitable for applications requiring high capacity over shorter distances.

10. Advantages of Sub-6 GHz :-

- Provides wider coverage.
- Better propagation over longer distances.
- Lower free-space path loss compared with higher frequencies.
- Better suitability for wide-area 5G deployment.
- Requires fewer base stations for a given coverage area.
- Provides good balance between coverage and capacity.

11. Advantages of mmWave :-

- Provides very high bandwidth.
- Supports extremely high data rates.
- Suitable for high-capacity applications.
- Useful for dense urban areas and hotspots.

- Suitable for applications requiring very low latency and high throughput.
- Can support large numbers of users in high-density locations.

12. Applications :-

Sub-6 GHz Applications

- Wide-area 5G mobile communication
- Rural and suburban coverage
- General mobile broadband
- IoT connectivity
- Vehicle communication
- Smart city communication

mmWave Applications

- 5G high-speed hotspots
- Stadiums and large public venues
- Dense urban areas
- Fixed wireless access
- High-speed indoor communication
- Industrial wireless communication

13. Limitations of the Simulation :-

The simulation uses a simplified Free-Space Path Loss model. Real-world wireless propagation is more complex.

The actual coverage area can be affected by:

- Buildings and obstacles
- Terrain
- Trees and vegetation
- Atmospheric conditions
- Rainfall
- Antenna height
- Antenna gain
- Beamforming
- Transmit power
- Receiver sensitivity
- Wall penetration losses

Therefore, the calculated coverage area should be considered a theoretical/model-based comparison rather than an exact real-world coverage value.

14. Conclusion :-

The simulation successfully compares the coverage area of **Sub-6 GHz and mmWave frequencies using Python**.

The results show that the **3.5 GHz Sub-6 GHz frequency provides a larger coverage range and coverage area** than the **28 GHz mmWave frequency** under the same transmission conditions.

The higher frequency of mmWave results in greater free-space path loss, causing the received signal power to decrease more rapidly as the distance increases.

Although mmWave provides significantly higher bandwidth and data-rate capabilities, its shorter coverage range makes it more suitable for dense and localized high-capacity applications.

Sub-6 GHz is therefore more suitable for **wide-area 5G coverage**, while mmWave is suitable for **high-capacity short-range communication**.

Overall, the Python-based simulation clearly demonstrates the fundamental **trade-off between coverage and frequency in 5G wireless communication systems**.